AI SYSTEM AUDIT REPORT

EU AI Act Compliance Assessment | Version 3.0.0

CONFIDENTIAL | ACADEMIC DRAFT

Executive Summary

This report presents a structured compliance audit of a machine learning-based credit risk classification system against the EU Artificial Intelligence Act (Regulation 2024/1689). The system is classified as High-Risk under Annex III, Point 5(b). The audit covers Articles 10, 12, 13, 14, and 15. All results were generated automatically by audit pipeline v3.0.0 on 2026-06-02.

Overall Risk Verdict

Overall Risk	Findings	Articles Reviewed	Deadline
HIGH	7 Identified	Arts. 10-15, 27	2 August 2026

Findings at a Glance

#	Finding	Article	Severity	Status
1	Gender Fairness (DIR Ratio)	Art. 10(2)(f)	**LOW**	**PASS**
2	Age-Based Discrimination Risk	Art. 10(3)	**HIGH**	**FAIL**
3	Missingness Bias - Checking Account	Art. 10(3)	**HIGH**	**FAIL**
4	Global Accuracy: 72.67%	Art. 15	**MEDIUM**	**ACCEPTABLE**
5	SHAP Explainability Map Produced	Art. 13	**LOW**	**PASS**
6	Human Oversight Not Documented	Art. 14	**HIGH**	**FAIL**
7	Continuous Risk Management Absent	Art. 9	**HIGH**	**FAIL**

Section 1 - System Identification

Field	Detail
System Name	German Credit Risk Assessment AI
Version	3.0.0
Classification	High-Risk - Annex III Point 5(b)
Algorithm	Random Forest Classifier
Dataset	German Credit Risk (UCI)
Audit Timestamp	2026-06-02 17:36:47
Articles Covered	Art.10(2), Art.10(3), Art.12, Art.13, Art.15
Deadline	2 August 2026

Section 2 - Audit Findings

Finding 1: Gender Fairness [LOW]

Field	Detail
Metric	DIR: 0.9969 | Female rate: 0.7917 | Male rate: 0.7941
Article	Art. 10(2)(f) - examination of possible biases affecting fundamental rights.
Result	Gender parity maintained above 4/5ths rule threshold of 0.80.

Finding 2: Age-Based Discrimination [HIGH]

Field	Detail
Metric	Young accuracy: 0.5745 | Adult accuracy: 0.7549 | DP Difference: 0.0829
Article	Art. 10(3) - training data shall be representative across subgroups.
Result	18-point accuracy gap. Young adults systematically disadvantaged.

Finding 3: Missingness Bias [HIGH]

Field	Detail
Metric	Top SHAP feature: Checking account_Missing = 0.0773
Article	Art. 10(3) - data shall be free of errors and complete.
Result	Missing data rewarded over financial merit. Governance failure.

Finding 4: Model Performance [MEDIUM]

Field	Detail
Metric	Accuracy: 0.7267 | Balanced: 0.6302 | Bad Risk Recall: 0.39
Article	Art. 15(1) - appropriate level of accuracy and robustness required.
Result	61% of bad risk borrowers incorrectly approved. Material financial risk.

Finding 5: Explainability [LOW]

Field	Detail
Metric	Top features: Checking account_Missing, Duration, Credit amount
Article	Art. 13(1) - transparency sufficient for interpreting outputs.
Result	SHAP analysis completed. Feature-level explanations available.

Finding 6: Human Oversight [HIGH]

Field	Detail
Metric	Human oversight documentation: NOT PRESENT
Article	Art. 14(1) - systems must allow effective human oversight.
Result	No override procedure or intervention mechanism documented.

Finding 7: Risk Management [HIGH]

Field	Detail
Metric	Risk management: ONE-TIME AUDIT ONLY
Article	Art. 9(1) - continuous iterative risk management required.
Result	No ongoing monitoring or lifecycle risk process in place.

Section 3 - Article 13 Transparency Map (SHAP)

SHAP values were computed for all test predictions. The chart shows the direction and magnitude of each feature's influence on credit decisions, satisfying Article 13(1) transparency requirements.

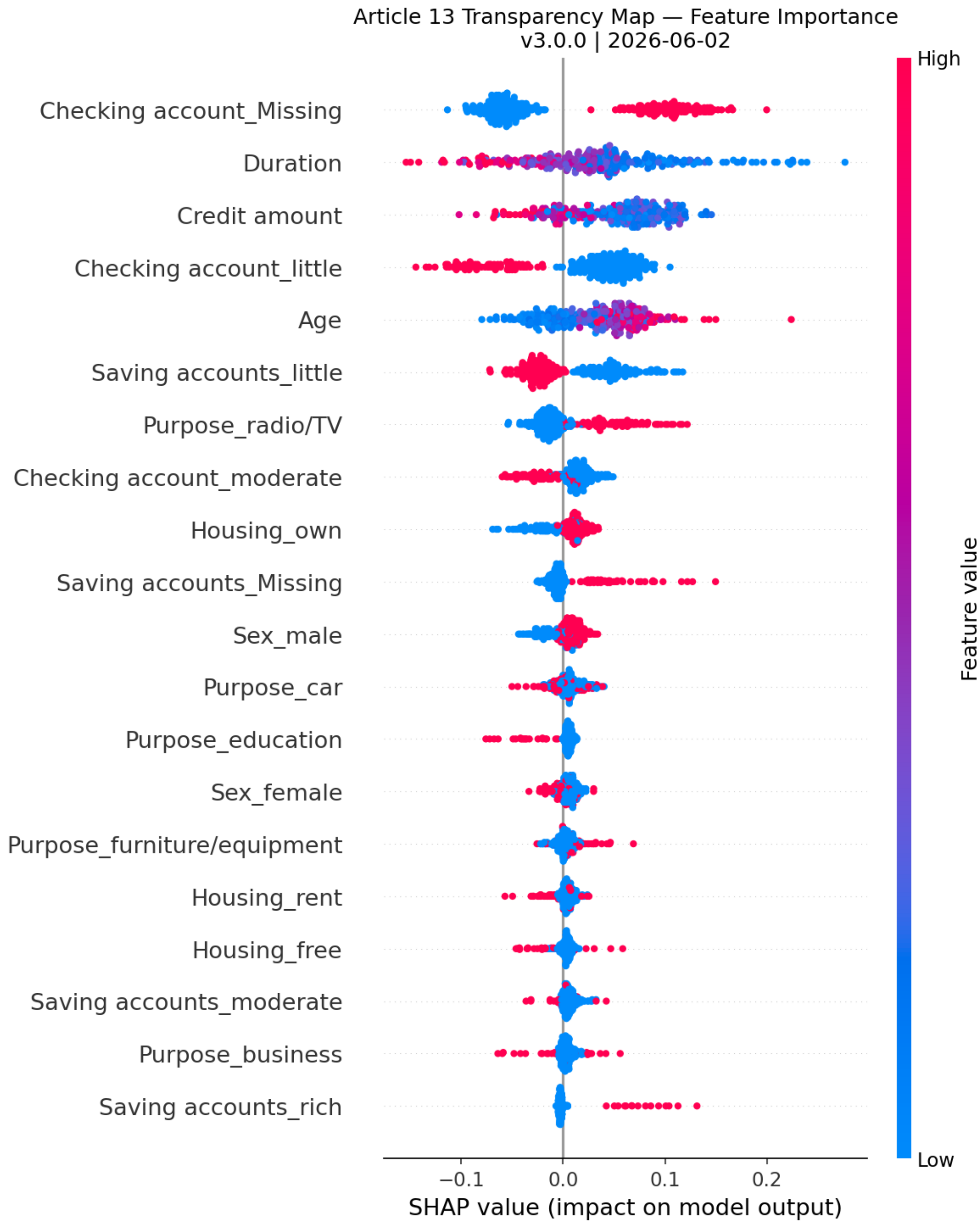

Figure 1: SHAP Summary Plot - Article 13 Transparency Map

Section 4 - Performance and Fairness Metrics

Model Performance

Metric	Value	Status
Global Accuracy	0.7267	ACCEPTABLE
Balanced Accuracy	0.6302	ACCEPTABLE
Bad Risk Recall	0.39	FAIL
Good Risk Recall	0.87	PASS
Test Set Size	300 records (30% split)	COVERED

Confusion Matrix

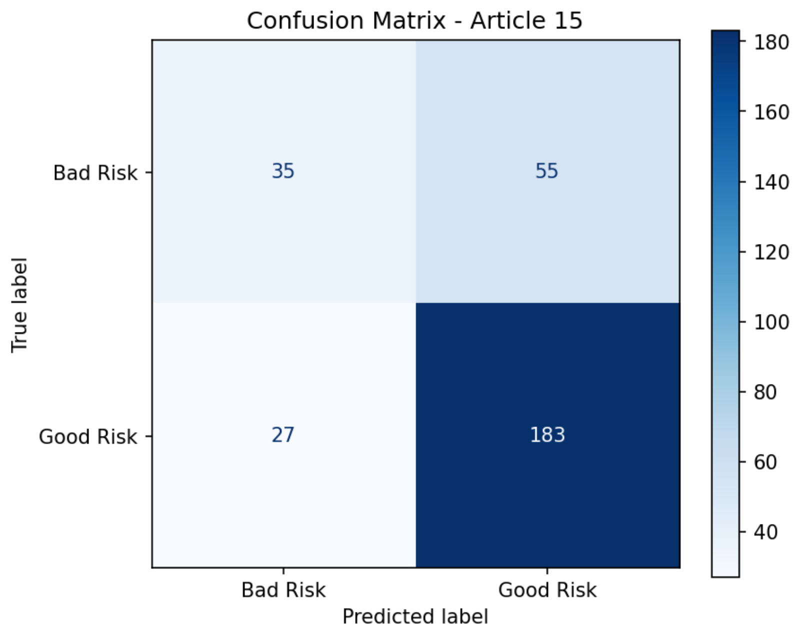

Figure 2: Confusion Matrix - Article 15 Robustness Check

Gender Fairness

Group	Accuracy	Selection Rate	Balanced Accuracy	Status
Female	0.6771	0.7917	0.5952	PASS
Male	0.7500	0.7941	0.6493	PASS

Age Group Fairness

Group	Accuracy	Selection Rate	Balanced Accuracy	Status
Young (<25)	0.5745	0.7234	0.5513	FAIL
Adult (25+)	0.7549	0.8063	0.6458	PASS

Fairness Visual Summary

The chart below visualises approval rates by demographic group against the 4/5ths rule threshold (0.80). Young adults clearly fall below the legal threshold, confirming the Article 10(2)(f) finding.

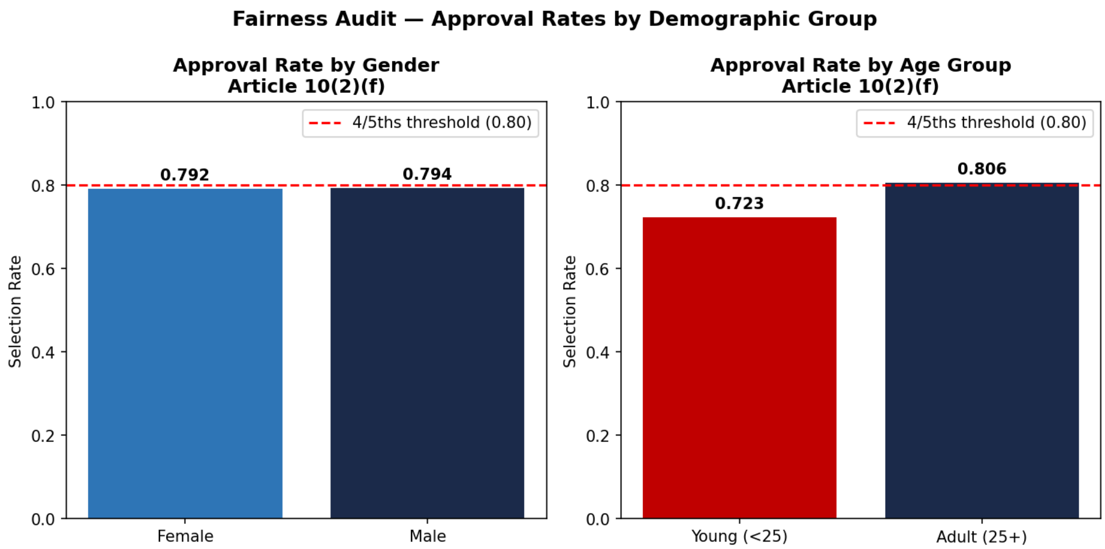

Figure 3: Fairness Audit — Approval Rates by Demographic Group

Section 5 - Threshold Justifications (Article 10(2))

Age Threshold

Field	Detail
Value	25
Legal Basis	Article 10(2)(f) EU AI Act
Justification	Age 25 chosen as young adult boundary based on ECB Financial Stability Review 2022 and Basel III retail exposure definitions. Individuals under 25 have structurally shorter credit histories due to systemic barriers, not lower creditworthiness. This is an auditing lens to detect proxy discrimination, not a model feature.

Section 8 - Human Oversight Checklist (Article 14)

Article 14 requires that high-risk AI systems allow effective human oversight during use. The checklist below documents the current compliance status of each requirement.

Requirement	Status	Risk	Recommendation
Art. 14(1) — Human oversight measures built into system	NOT IMPLEMENTED	**HIGH**	Define a mandatory human review process for borderline cases where model confide
Art. 14(3)(a) — Humans can override automated decisions	NOT IMPLEMENTED	**HIGH**	Implement an override flag in the decision output allowing loan officers to reve
Art. 14(3)(b) — System can be monitored during operation	**PARTIAL**	**MEDIUM**	Audit log exists but no real-time monitoring dashboard is in place. Add drift de
Art. 14(3)(e) — System can be stopped if risk identified	NOT IMPLEMENTED	**HIGH**	Document a formal shutdown procedure and assign responsibility to a named role.
Art. 14(4) — Deployers have sufficient AI literacy	NOT IMPLEMENTED	**MEDIUM**	Provide training documentation for staff who interact with or rely on model outp

Test Size

Field	Detail
Value	0.3
Legal Basis	Article 15 EU AI Act — robustness
Justification	30% test split chosen over 20% because with 1000 samples, a 300-sample test set ensures subgroups (young females) have n>30 for statistically reliable fairness metrics.

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Dir Threshold

Field	Detail
Value	0.8
Legal Basis	Article 10(5) EU AI Act — bias mitigation
Justification	The 4/5ths rule (0.80 Disparate Impact Ratio) is the internationally recognised standard from EEOC Uniform Guidelines and referenced in NIST AI RMF. Below 0.80 = prima facie discrimination.

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Section 6 - Remediation Roadmap

#	Action Required	Article	Priority	Target
1	Document human oversight and override mechanism	Art. 14	**CRITICAL**	Pre-deployment
2	Remediate missingness bias in checking account feature	Art. 10(3)	**CRITICAL**	Pre-deployment
3	Establish continuous risk management process	Art. 9	**HIGH**	Pre-deployment
4	Improve bad risk recall via resampling or threshold	Art. 15	**HIGH**	30 days

	tuning			
5	Address age fairness gap - young adult accuracy 0.57	Art. 10(2)(f)	**HIGH**	30 days
6	Conduct Fundamental Rights Impact Assessment	Art. 27	**MEDIUM**	60 days

Section 7 - Audit Trail (Article 12)

Field	Detail
Timestamp	2026-06-02 17:36:47
Pipeline Version	3.0.0
Log File	outputs/logs/audit_log_v3.0.0.json
Seed	42 (fixed for reproducibility)
Preprocessor	ColumnTransformer (StandardScaler + OneHotEncoder)
Classifier	RandomForestClassifier (class_weight=balanced)

Glossary

Term	Definition
High-Risk AI	AI system listed in Annex III subject to mandatory compliance.
DIR	Disparate Impact Ratio. Below 0.80 = prima facie discrimination.
SHAP	SHapley Additive exPlanations. Explains individual predictions.
Balanced Accuracy	Average recall across classes. Robust for imbalanced datasets.
DP Difference	Demographic Parity Difference. 0 = perfect parity between groups.
FRIA	Fundamental Rights Impact Assessment. Required under Art. 27.